

KEIT, One Year On:

Measuring Kubernetes Carbon on a Sovereign Cloud

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eerste dag van uitgifte 2013

first day of issue 2013

Dag van de postzegel

EERSTE DAG
VAN UITGIFTE
DEN HAAG
18 OKTOBER
2013

EERSTE DAG
VAN UITGIFTE
DEN HAAG
18 OKTOBER
2013



dag van de
postzegel
2013



NR

682



FOSDEM



H-2214

Thank you! Questions?

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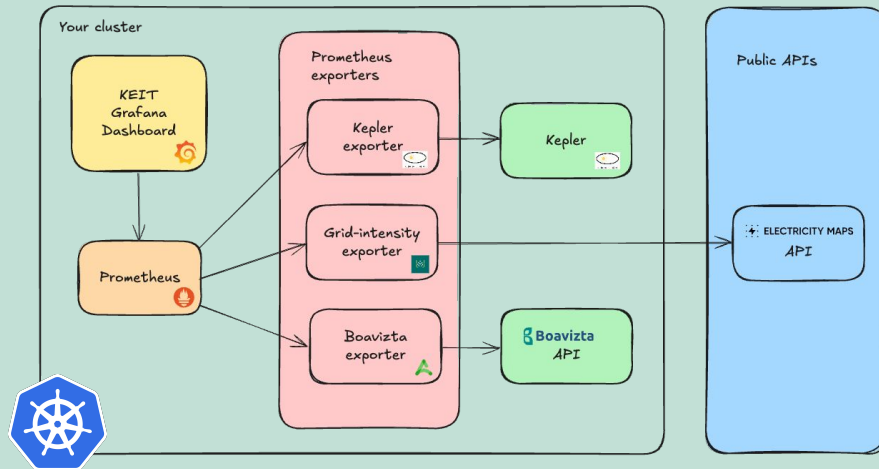
flavia@aknastic.com



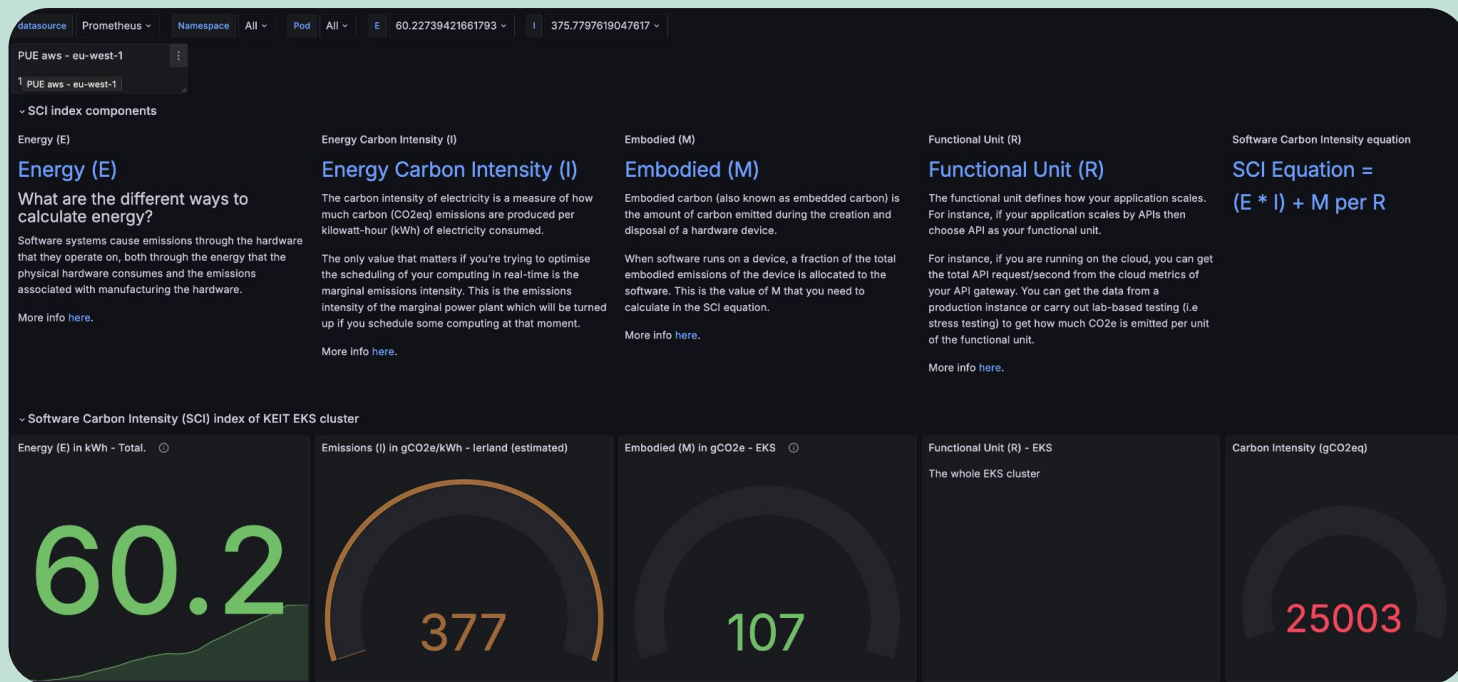
#2025-
#energyd

What's KEIT? (Kubernetes Emissions Insights Tool)

- Goal: expose emissions in Kubernetes clusters
- Computed as SCI per the Green Software Foundation spec
- Components:
 - Kepler (energy)
 - ElectricityMaps (intensity)
 - Boavizta API (embodied)
- Output: per-pod, per-namespace SCI on a live dashboard
- Bare metal and AWS-enabled



Insights from KEIT – the dashboard



What changed in the past year?

**For one, it became apparent that it's
dangerous to depend fully on US-American
big tech**



Criminal Court: Microsoft's email block a wake-up call for digital sovereignty

Microsoft blocked the email account of Chief Prosecutor of the International Court of Justice after Trump's sanctions. Critics: "We urgently need alternatives."

May 19, 2025 at 8:52 am CEST 4 min. read

<https://www.heise.de>

The Cloud Act

Microsoft admits it 'cannot guarantee' data sovereignty

Under oath in French Senate, exec says it would be compelled – however unlikely – to pass local customer info to US admin

 Paul Kunert

100 

Published Fri 25 Jul 2025 // 13:00 UTC

<https://www.theregister.com/>



T Tech Times

EU Tech Sovereignty Package Debuts as CISA's Own Cloud Keys Sat Exposed for Months

EU Tech Sovereignty Package restricts AWS, Azure, and Google Cloud from sensitive EU government data — on the same day a Nightwing...

2 days ago



The reaction



Biometric Update

Dutch court backs DigiD contract renewal amid U.S. CLOUD Act fears

Sovereignty debates are emerging across Europe as governments reassess dependence on foreign-owned cloud, identity and public-sector...

1 week ago



What changed in the past year?



We realised our data and core digital systems are at the whim of dangerous governments

**Cloud
Sovereignty**



The ground shifted under us... no more support for VMs..

**Kepler
refactoring**



Some European clouds provide carbon APIs

**Native Carbon
APIs**



How about comparing **KEIT**
measurements with **Scaleway's**
Environmental Footprint API?



Our setup: Scaleway sovereign cloud

- Running on Scaleway Kapsule
- 6 projects running
- Chose “Groupware” cluster where we were running GitLab, Nextcloud, Mattermost, Plausible to install KEIT/Kepler
- Running in France region
- On 3 nodes

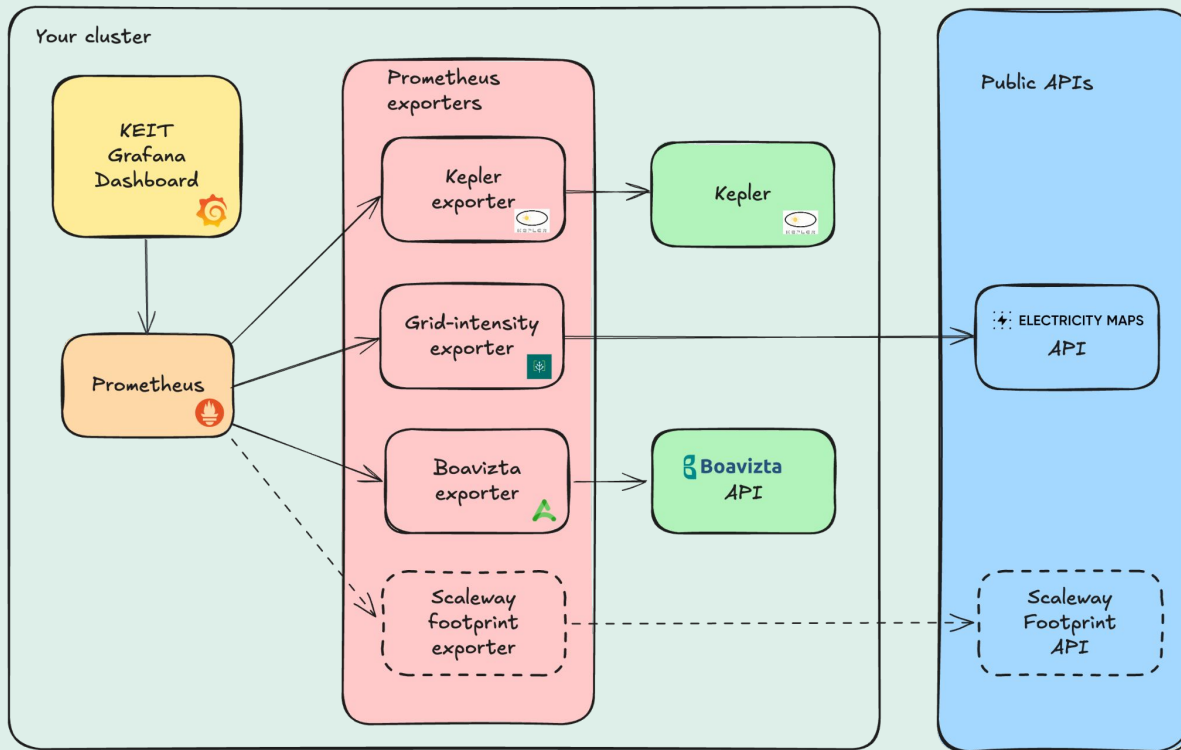


Scaleway Environmental Footprint console dashboard

Extended KEIT to import Scaleway data

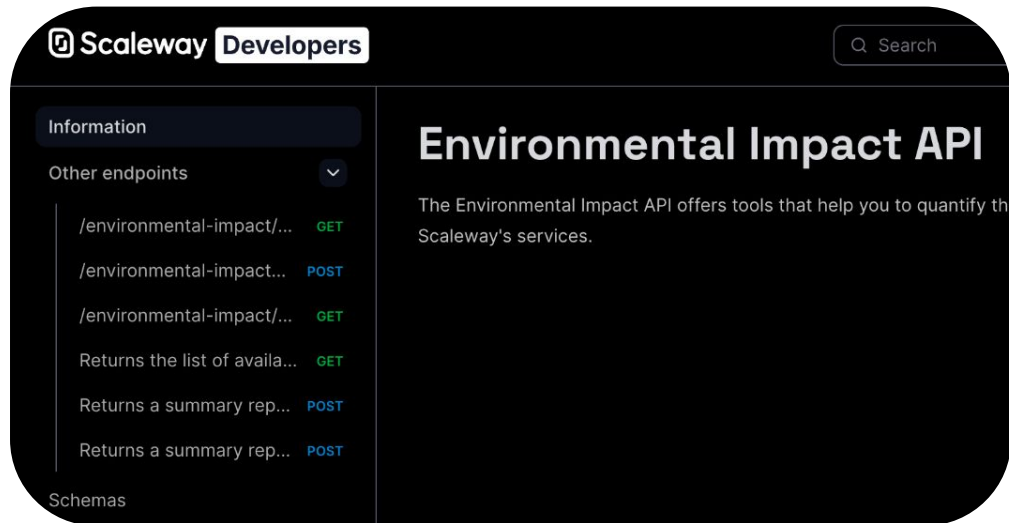
Created Scaleway
Prometheus
exporter:

- scraping hourly
- 14-day rolling window



What we learned from the Scaleway API

- Data sometimes lags 3–4 days (solved with a rolling 14-day window)
- SKU is a resource type; per-pod or namespace attribution not possible
- Scaleway has announced an August 2026 methodology refresh — the numbers will shift



All three of these are findings I wouldn't have predicted from reading the docs.

Then we compared (KEIT vs Scaleway API)

For 5 days:

- KEIT-side SCI:

2 kgCO₂eq

- Scaleway-side bundled:

7 kgCO₂eq

- Ratio: **3.5× !!!**

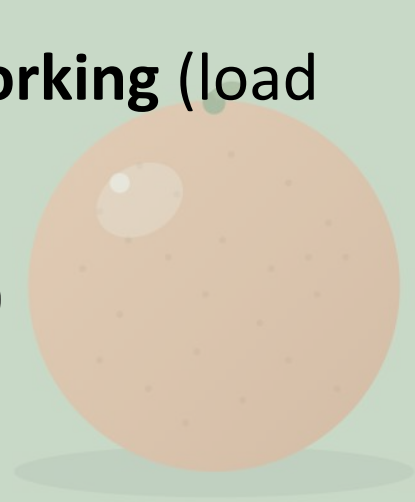
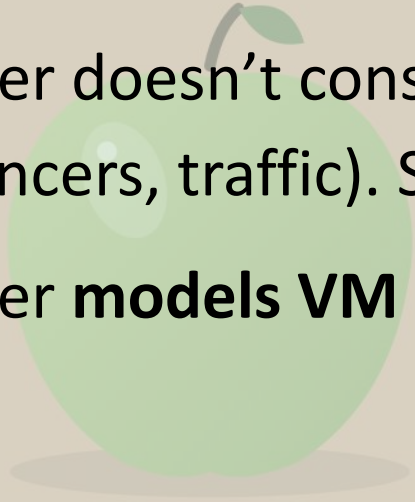


Why such a big difference??



Why such a big difference?

- Scaleway gives result for **the whole project: 2 clusters**. KEIT was installed on one
- Kepler doesn't consider **storage, networking** (load balancers, traffic). Scaleway does.
- Kepler **models VM** energy (old version)



**Each tool is measuring
different things**



Checking Kepler's logs...

```
Kepler running on version: v0.8.0
failed to open path /dev/cpu/0/msr: no such file or directory    ← no RAPL
Unable to obtain power, use estimate method                      ← fallback
Could not find any ACPI power meter path. Is it a VM?          ← Kepler asks itself
using none to obtain power
Using the Ratio Power Model to estimate PROCESS_TOTAL Power    ← regression model
Feature names: [bpf_cpu_time_ms]                               ← single feature
Requesting for Machine Spec: amd_epyc_7543_32_core ...         ← hardcoded CPU profile
```

We confirm that Kepler isn't measuring on
Kapsule (RAPL). It's modelling.

The Kepler rewrite

- In July 2025, Kepler 0.9.x was frozen, "no more updates"
- Soon after, there was a rewrite: v0.10.0, which drops legacy power models
- In Feb 2026 VM support still not restored
- Today this is not supported, but it's being investigated

 **Unfortunately, the most popular K8s energy tool doesn't support VMs any more, which is where most users run.**

But some maintainers are looking into this...

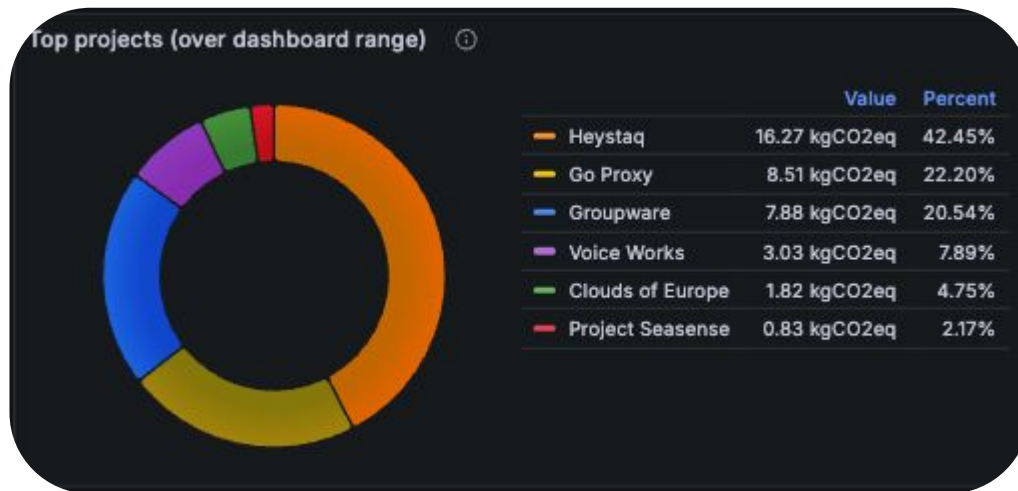
- Is it all bad?
- Not at all!!

Let's check out our findings...



What looking at Scaleway showed us

- **Observability** is our biggest emitter (Heystaq)



What looking at Scaleway showed us

- Compute and load balancer in **Warsaw** stand out

▼ The regional grid effect (often the biggest lever)

Same machine, different zones — heatmap (red = more emissions) ⓘ

sku_type\zone	fr-par-2	fr-par-3	fr-par-1	nl-ams-1	nl-ams-2	pl-waw-1	pl-waw-2
/compute/pro2_xs	0.29 kgCO2eq	0.77 kgCO2eq	0.24 kgCO2eq				
/compute/dev1_xl			0.45 kgCO2eq				
/compute/pro2_xxs	1.00 kgCO2eq		0.79 kgCO2eq				
/compute/dev1_m	0.34 kgCO2eq		0.55 kgCO2eq	1.54 kgCO2eq	3.86 kgCO2eq	11.66 kgCO2eq	5.26 kgCO2eq
/compute/pro2_s	1.78 kgCO2eq	3.79 kgCO2eq	1.58 kgCO2eq				
/network/loadbalancer			1.89 kgCO2eq	1.39 kgCO2eq		5.57 kgCO2eq	

What looking at Scaleway showed us

- **Storage** strangely stood out as a big emitter, very close to compute



We found buckets with unwanted terabytes of data

The screenshot shows the AWS Object Storage console interface. At the top, there are tabs for 'Buckets' (26) and 'Bucket policies' (5). Below this, a summary bar shows 'Data usage: 2.77 TB', 'Total objects: 157,977', and 'Total buckets: 26'. A table lists the buckets, with columns for Name, Visibility, Policy, Objects, Size, and Region. Three buckets are highlighted with red circles around their 'Size' values: 'groupware-postgres-backups' (2.57 TB), 'heystaq-mimir-blocks-v5' (2.35 TB), and 'heystaq-tempo-traces-v5' (1.44 TB).

Object Storage

Buckets **26** Bucket policies **5** NEW

Your Object Storage usage

PAR > **Data usage: 2.77 TB** **Total objects: 157,977** **Total buckets: 26**

Buckets are groups of objects sharing a common denominator. You can store as many objects as you want in each bucket. [Read the Object Storage quickstart.](#) [+ Create a Bucket](#)

☐	Name	Visibility	Policy	Objects	Size	Region	
☐	groupware-postgres-backups	Private	No policy	188,610	2.57 TB	PAR	...
☐	heystaq-mimir-blocks-v5	Private	No policy	241,596	2.35 TB	PAR	...
☐	heystaq-tempo-traces-v5	Private	No policy	168,448	1.44 TB	PAR	...

Spotted, diagnosed, fixed

- We found buckets with terabytes of data which we didn't expect
- It turned out that we had **no lifecycle policy** for these buckets and **versioning was enabled**
- We fixed it in a few days and saved **hundreds of kg CO2 per year**



Look, optimise, check next hotspot

Look



You can't
improve
what you
don't
measure



Optimise



**Precise measurements are not
necessary to make improvements**



What you can still do with all this

- Relative comparisons
- Find hotspots
- Regression detection
- Triangulation as sanity check — when two methodologies agree on a shape, trust it more

Disclaimer: using GenAI for this work

- Good for:
 - A thinking partner, from the what to the how
 - Building Grafana dashboards from scratch
 - Analysing API responses and data
- Not good for:
 - Judging if findings are actually useful
 - "What story does this dashboard need to tell?"
 - Pivoting when encountering unexpected results
- AI shortened the path from data to question. **The questions still need a human.**
- The **irony** of using a carbon-heavy tool to investigate carbon vs **AI for sustainability**

So what to do next?

Start **looking** - you'll find more than you think

Use **tools that are already out there**, either from your provider, open source, etc.

Find hotspot → improve → find next hotspot

Incorporate best practices for cloud native sustainability in your daily work

If you want more: **contribute back** to the open-source tools you use.



And share it!

Useful links

- Everything about this talk, including slides
github.com/aknastic/green-io-amsterdam-2026
- KEIT repository github.com/aknastic/keit
- [Best practices for cloud native sustainability](#)
- [Scaleway Environmental Footprint API](#)
- Kepler repository github.com/sustainable-computing-io/kepler
- Join the CNCF Sustainability TCG in [Slack](#)

Thank you! Questions?



[Flavia @ Aknostic.com](mailto:Flavia@Aknostic.com)

KEIT: github.com/aknostic/keit

Feedback

